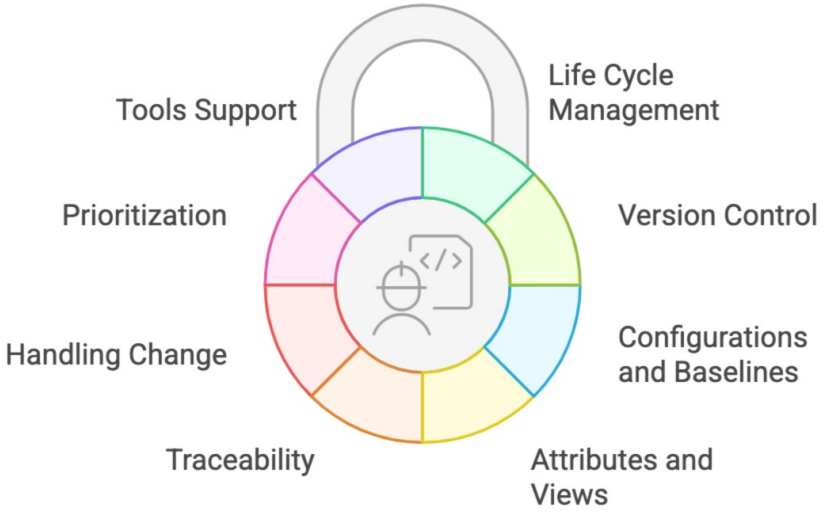
SCSE2123 SOFTWARE REQUIREMENTS ENGINEERING

TOPIC 8: Practices for Requirements Management

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CONTENTS



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Objectives

Students will be able to:

- **explain** what requirements management is about and why is it needed
- **explain** the purpose of attributes set for requirements
- **explain** the purpose of views and name the different views of requirements
- **explain** what a requirements versioning concept looks like in a given project situation
- **analyze** the requirements changes in linear (plan-based) and agile approaches
- **justify** the steps and reason for prioritization based on the meaningful assessment criteria

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Terms:

Requirements management

Change management

Traceability

Requirements attributes

Requirements life cycle

Prioritization

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WHAT IS REQUIREMENTS MANAGEMENT?

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Definition: Requirements Management

The process of managing existing requirements and requirements-related work products, including the storing, changing, and tracing of requirements (↑ traceability).

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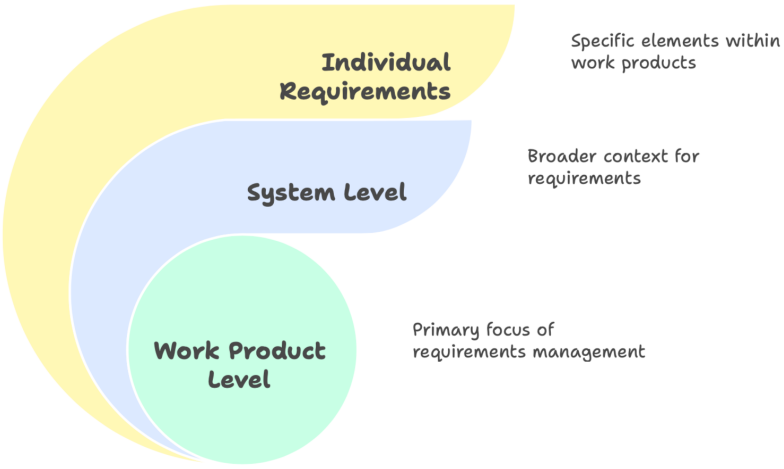
Requirements Management: What

- **Activities** that **identify**, **document**, **maintain**, **communicate**, **trace** and **track requirements** throughout the **life cycle** of a **system**, **product** or **service**. (*from ISO/IEC/IEEE 29148:2018*)
- Requirements Engineer manages requirements so that all roles in a project can work effectively and efficiently.

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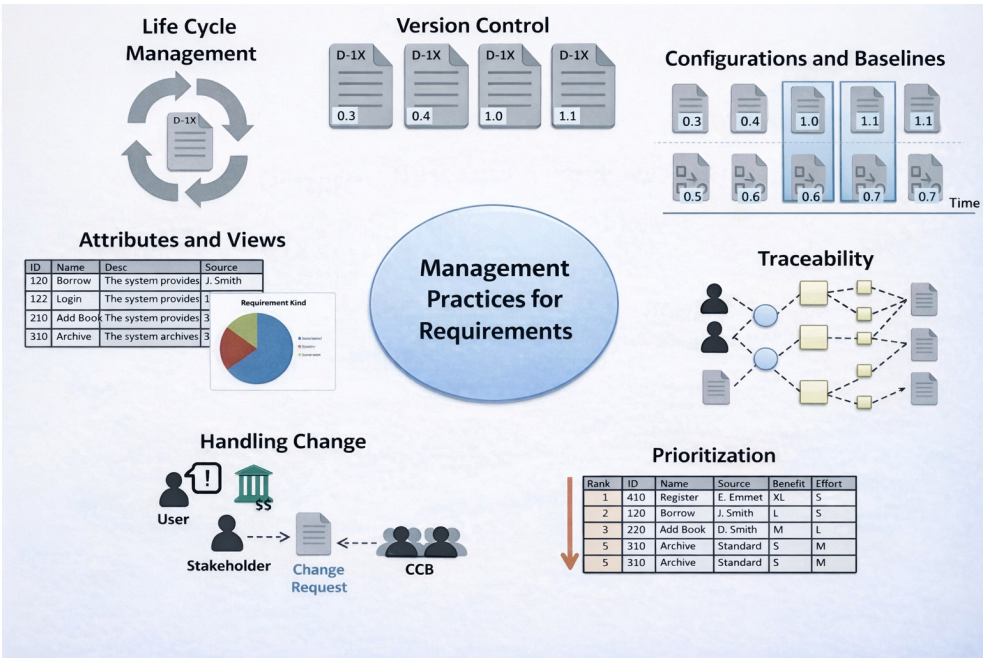
7

… Different Levels of Requirements Management



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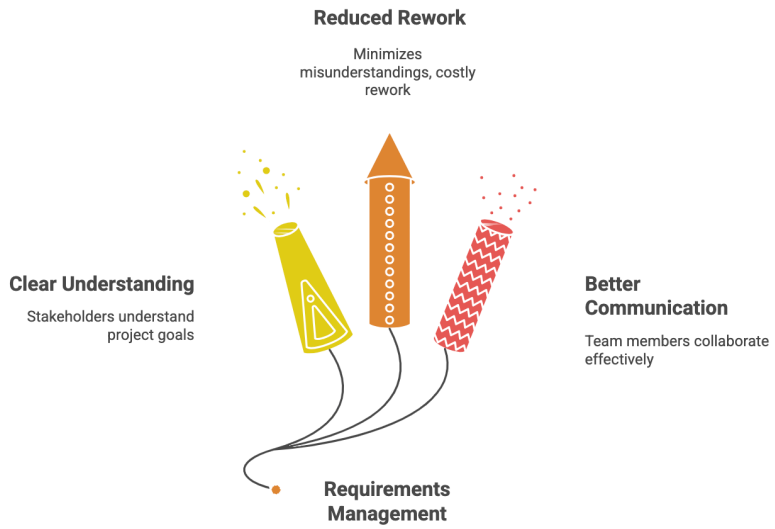
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Source: Mathias Lampe (2024)

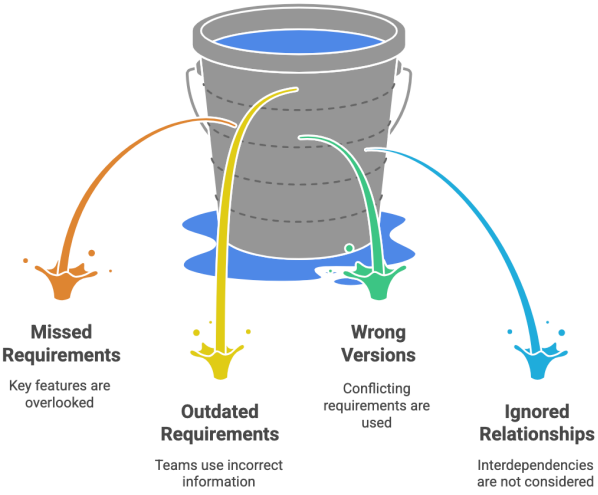
Benefits of Requirements Management

- Improve efficiency and effectiveness of system development and usage



Problems When Requirements Are Not Managed

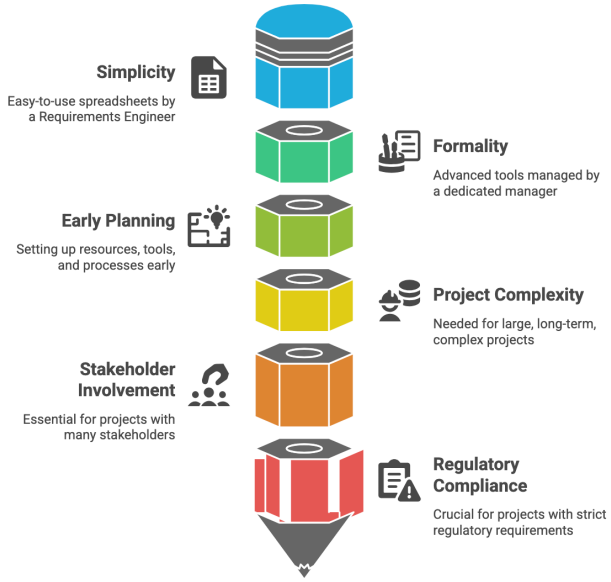
- Overall system development becomes inefficient and ineffective



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Requirements Management (cont.)



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Scenarios Examples

Requirements Management Complexity and Criticality

Healthcare systems

Healthcare systems prioritize high criticality with relatively low complexity.



Banking systems

Banking systems require high complexity and criticality due to strict regulations.



Government systems

Government systems involve low complexity and criticality with many stakeholders.



Large enterprise systems

Large enterprise systems face high complexity but lower criticality.



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LIFE-CYCLE MANAGEMENT

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Requirements Life Cycle Management

1

Tracks requirement status through life cycle

- Tracks the status of requirements and work products throughout their life cycle
- Each requirement/work product moves through states (e.g. created, reviewed, approved)

2

Life cycle model defines statuses and transitions

- A life cycle model defines:
- allowed statuses
- valid transitions between statuses

3

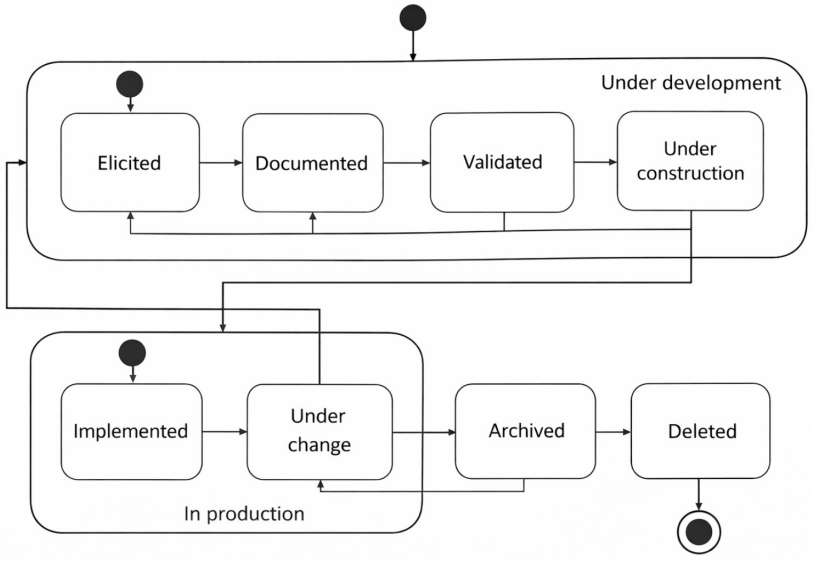
Current status and history must be clear

- The current status must always be clear, including its change history

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Example: Life Cycle of a Requirement



(from Handbook for the CPRE Foundation Level according to the IREB Standard, v 1.0, 2020)

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Life Cycle Management – Complicating Factors

Overlapping Life Cycles

Requirements and work products have different, overlapping life cycles. A work product being under change does not mean all requirements in it must be changed.

Different roles focus on different life cycle states. Requirements Engineer, Project Manager, Product Manager, Change Manager focus on different states.

Different Role Focus

Requirement States

Requirement states must be actively tracked and clearly recorded. Past, current, and future state of requirements, how they can change, and why.

State of a work product is often recorded as an attribute. This helps in understanding the current status and history of the product.

Work Product Attributes

Tailored Management

Life cycle management should be tailored to project and customer needs. This ensures the process is efficient and effective for the specific context.

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VERSION CONTROL

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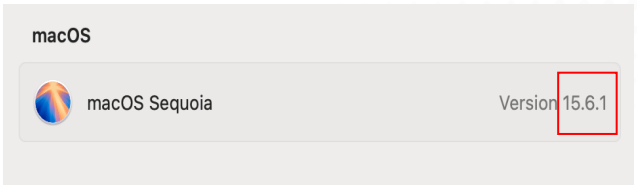
Definition: Version

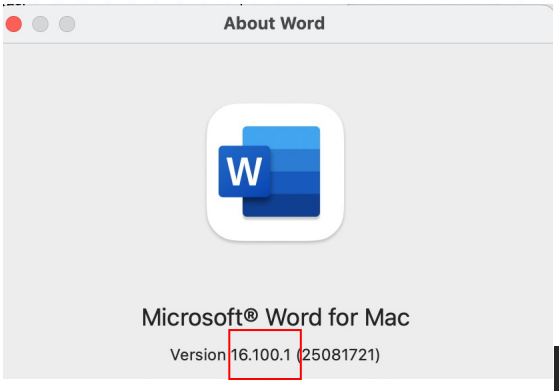
An occurrence of an ↑item which exists in multiple, time-ordered occurrences where each occurrence has been created by modifying one of its previous occurrences.

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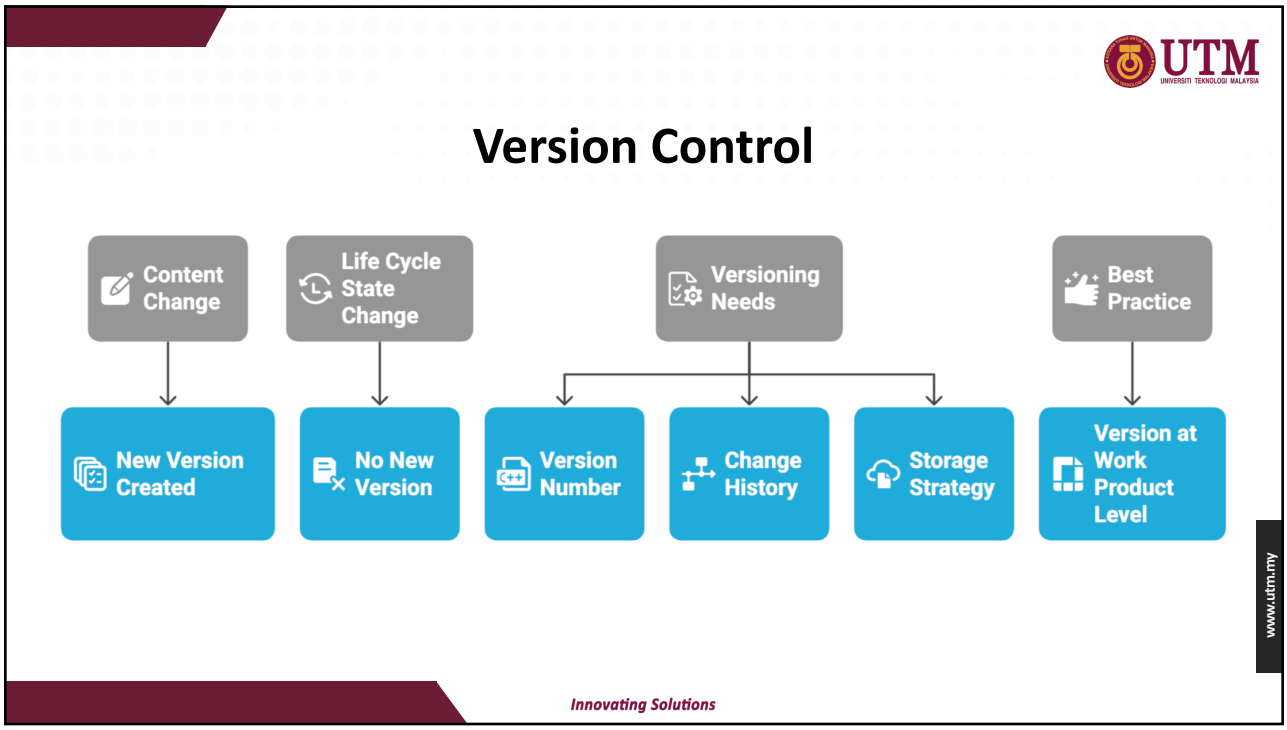
Example



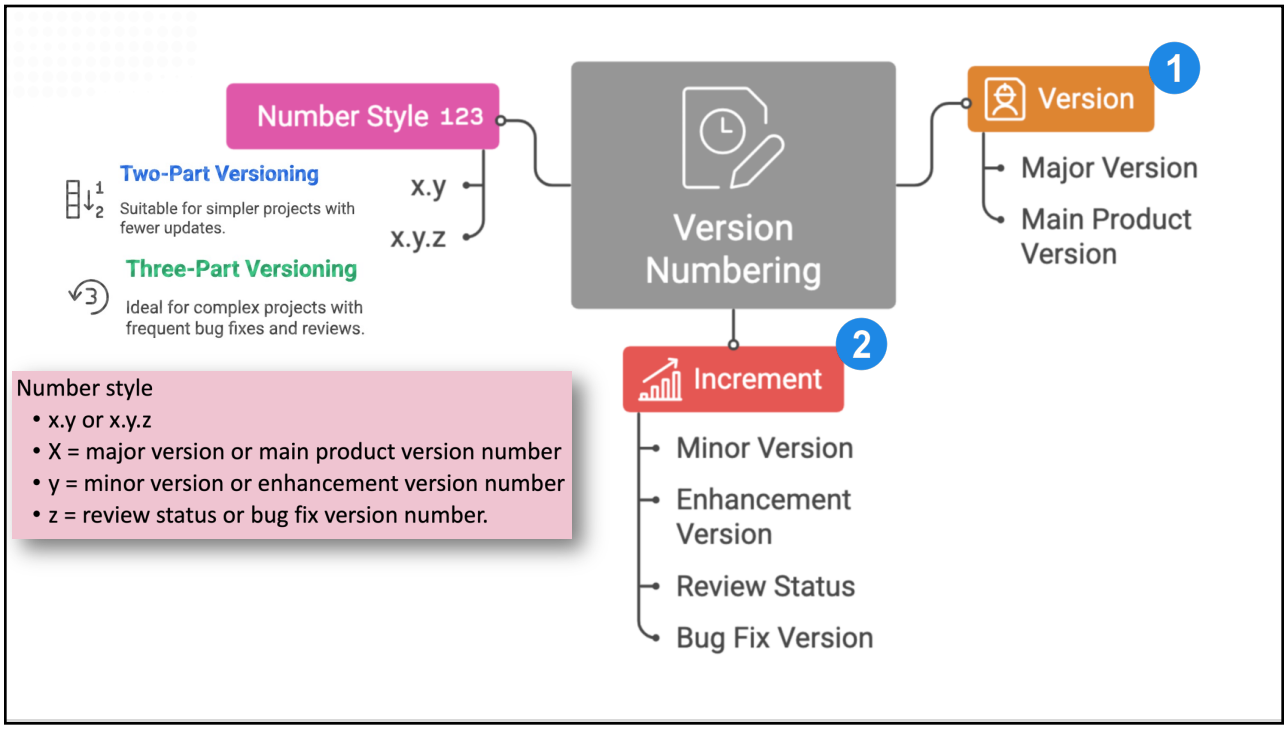


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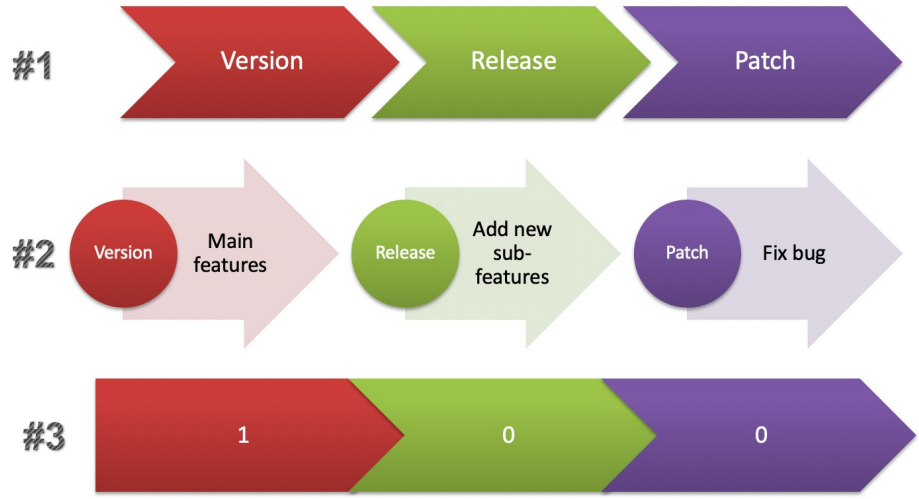

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Version Control
Content Change
Life Cycle State Change
Versioning Needs
Best Practice
New Version Created
No New Version
Version Number
Change History
Storage Strategy
Version at Work Product Level
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Number Style 123
Two-Part Versioning
Suitable for simpler projects with fewer updates.
x.y
x.y.z
Three-Part Versioning
Ideal for complex projects with frequent bug fixes and reviews.
Version Numbering
1
Version
Major Version
Main Product Version
2
Increment
Minor Version
Enhancement Version
Review Status
Bug Fix Version
Number style
• x.y or x.y.z
• X = major version or main product version number
• y = minor version or enhancement version number
• z = review status or bug fix version number.

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Version Number Basic Concepts

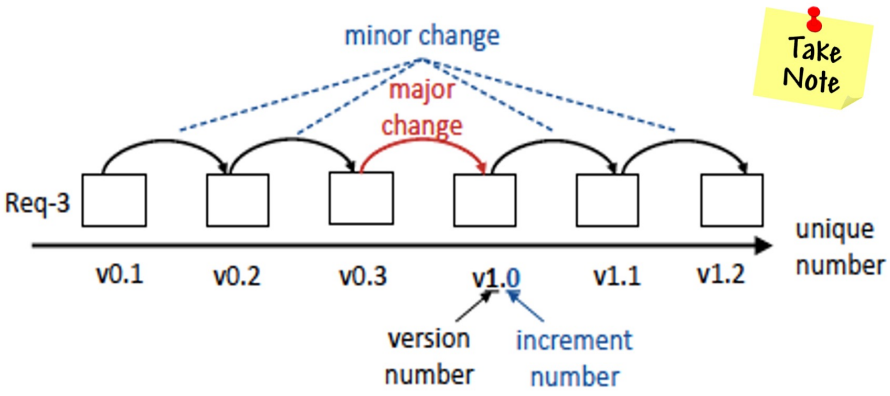


Example of Version number : v1.0.0

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Version Numbering – Software Feature



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Version Numbering – Software Feature: Example

Major Enhancement (X.y.z)

- A Major enhancement involves significant changes that often alter the core functionality or structure of the work product.
- This typically corresponds to a change in the first digit (X) of the version number.
- Main product version or significant overhaul
- Example: Introducing a completely new module, such as adding a "Payment Gateway" to an e-commerce app that previously only supported "Cash on Delivery."
  - Version Change: 1.0.0 → 2.0.0

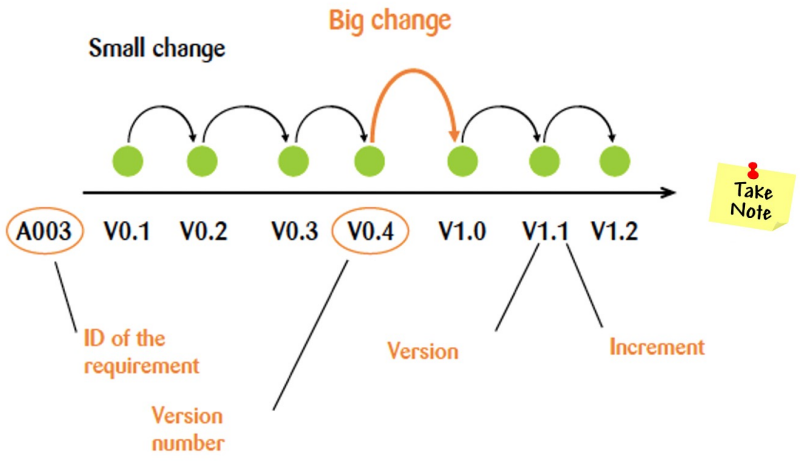
Minor Enhancement (x.Y.z)

- A Minor Enhancement involves smaller additions or improvements that do not fundamentally change the system's core purpose.
- This corresponds to a change in the second digit (Y) of the version number.
- Enhancement or small feature addition
- Example: Adding a "Filter by Color" option to an existing search feature in an e-commerce app.
  - Version Change: 1.2.0 → 1.3.0

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Version Numbering – Document Revision



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Version Numbering – Document Revision: Example

Major Enhancement (X.y.z)

- A Major enhancement involves significant changes that often alter the core functionality or structure of the work product.
- This typically corresponds to a change in the first digit (X) of the version number.
- Main product version or significant overhaul

- Example: Rewriting the entire "System Architecture" section of a Requirements Specification document to support a cloud-based infrastructure instead of an on-premise server.
  - Version Change: 2.4.0 → 3.0.0

Minor Enhancement (x.Y.z)

- A Minor Enhancement involves smaller additions or improvements that do not fundamentally change the system's core purpose.
- This corresponds to a change in the second digit (Y) of the version number.
- Enhancement or small feature addition

- Example: Adding a new subsection to an existing chapter, such as defining two new "User Roles" in the Stakeholder Requirements section.
  - Version Change: 1.2.5 →1.3.0

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Control Change Document for Versions

Version	Descriptions (Comments)	Delivered Date	Associated files name

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Control Change Document for Versions: Example of Iman Apps.

Version	Descriptions (Comments)	Delivered Date	Associated files name
1.0	Size: 1.5 MB Language: English Seller: Adiman Muhammad	Mar 09, 2012	http://iman.adiman.net/iman-1-0-launched/
1.2.2	1. FIXED issue where prayer time not showing on some devices. - FIXED bug that causes iman to crash when connection to JAKIM server not available. 2. Iman will now display last known location data from JAKIM whenever failure to fetch new data from the server 3. Added refresh button on Prayer Time page. Easier for user to reload prayer time data instead going into the app preferences 4. Added adzan notifications, with customization to toggle on/off for each prayer time	Mar 26, 2012	http://iman.adiman.net/iman-1-2-2-is-now-ready-for-download-in-appstore/
2.0	1. New compass user interface 2. New features Jama Qasr' english translation!	May 26, 2012	http://iman.adiman.net/iman-2-0/
…	1. …..	….	….

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Project P3 – SRS Document Enhancement

How is versioning **applied** in our class project?

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Project PR3 – SRS Document Enhancement

Topic 8: Versioning

History Log:	*</ID>*	*<Changes>*
Version:		
Use case ID:		
Use case name:		
Created by:		**Last Updated by:**
Date Created:		**Last Revision Date:**
Description:		
Actor(s)		
Pre-condition:		
Normal Flow(s):		
Alternative Flow(s):		
Exceptions Flow(s):		
Post-condition:		
Related Requirement(s):	**ID**	**Requirement** / **Priority**

Topic 8: Attributes assignment & Traceability

Topic 8: Attributes assignment & Traceability

Topic 8: Attributes assignment & Traceability

Topic 8: Prioritization (using Kano)

TOPIC 6 : Write single requirement using template/pattern
+ TOPIC 5 : quality criteria consideration

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3.2.2.1 UC-005: Issue Invoice

History Log:	1.0	Create initial use case
	2.0	Fixed alternate flow
		Fixed exception flow
		Fixed normal flow
	3.0	Fixed pre-conditions
		Fixed normal flow
		Added related requirements
Version:	3.0	
Use Case ID :	UC-005	
Use Case Name :	Issue Invoice	
Description:	This use case describes how the Pusat Pengumpulan OC and Pusat Timbang Setempat OC issue invoice.	
Created By:	Fatin Nabillah bt Sihan	**Last Updated By:** Pang Cheng Hao
Date Created:	11 April 2016	**Last Revision Date:** 15 May 2016
Actors :	Pusat Pengumpulan(PP), Pusat Timbang Setempat(PTS) OC.	
Description :	This use case describes how the PP and PTS creates invoice.	
Pre-conditions :	1. The PP and PTS OC must first login into the plantation management system. 2. The PP and PTS OC must receive weighing ticket. 3. The PP and PTS OC clicks "Invoice" button.	
Normal Flow :	1. The PP and PTS OC enters the buyer's company name. 2. The PP and PTS OC enters the price, weight and grades based on weighing ticket. 3. The system auto generates date and time for the invoice. 4. The PP and PTS OC clicks "Save" button. (See A.1) 5. The system saves the invoice in system database. 6. The PP and PTS OC clicks "Print" button. 7. The system prints the invoice (See E.1)	
Alternative Flow :	A.1 **The PP and PTS OC cancel invoice** 1. The PP and PTS OC clicks "Cancel" button. 2. Return to normal flow step 1. (NF 1.0)	
Exception :	E.1 **The system fails to print.**	

Example of improvised UC specification with enhanced attributes @ *added information*

Disclaimer:
The version ID history logs were not correctly specified. Please revise appropriately.

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	1. The system display "Error in printing process. Please check printer." 2. Return to normal flow step 7. (See NF 1.0)		
Post-conditions :	The PP and PTS OC successfully created invoice and save into database		
Related requirement:	**ID**	**Requirement**	**Priority**
	FR UC-005-01	The system shall be able to create an invoice	**Basic**
	FR UC-005-02	The system shall be able to display the Invoice form.	**Basic**
	FR UC-005-03	The system shall be able to allow PP and PTS OC to enter details into the Invoice form	**Basic**
	FR UC-005-04	The system shall allow PP and PTS OC to print the created Invoice	**Excitement**
	FR UC-005-05	The PP and PTS OC should be able to cancel the Invoice	**Basic**
	QR UC-005-06	The system shall be able to prompt error message when system fails to print the Invoice.	**Performance**
	CR UC-005-07	The PP and PTS OC shall be able to print the Invoice after the it has been successfully saved in system database	**Performance**

Table 4: Use case description for UC-005

Next Sub-topic: Prioritization (using Kano technique)

Uniquely assign the requirements ID based on types
i.e. FR – functional |
QR = quality |
CR = constraint

Note:
Requirements types covered in Topic 1

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History Log:	**0.1** **…**			
Version:	**<Latest version no.>**			
Misuse Case ID:	<unique ID of the misuse case>			
Misuse Case Name:	<A short active verb phrase>			
Created By:	**<Team PIC>**		**Last Updated By:**	**<Team PIC>**
Date Created:	**<dd/mm/yyyy>**		**Last Revision Date:**	**<dd/mm/yyyy>**
Misuser(s)/Threat Agent(s):	<Name of the unauthorized actor or external agent involved>			
Description:	<Short description of the security threat or negative event>			
Threat:	<The security vulnerability or flaw this misuse case exploits>			
Triggers	<Event that initiates the execution of the misuse case (the attack start)>			
Pre-Condition(s):	<Condition(s) that must be true before the misuse case can be executed (e.g., *System is online*, *Threat agent has network access*)>			
Misuse Flow (Attack Steps):	<Description of the steps the misuser takes to execute the attack/harm>			
Mitigation Flow(s):	<Description of alternative or option steps that could be taken by the system (or a legitimate user) to stop the attack mid-flow>			
Exception Flow(s):	<Description of how the system handles critical errors during the attack and what prevents the attack from succeeding>			
Post-conditions:	<The resulting state of the system if the misuse case runs successfully (the harm done)>			
Related Use Case(s):	<The legitimate use case(s) this misuse case threatens or is the inverse of>			
Security Requirement:	**ID**	**Requirement**		**Priority**

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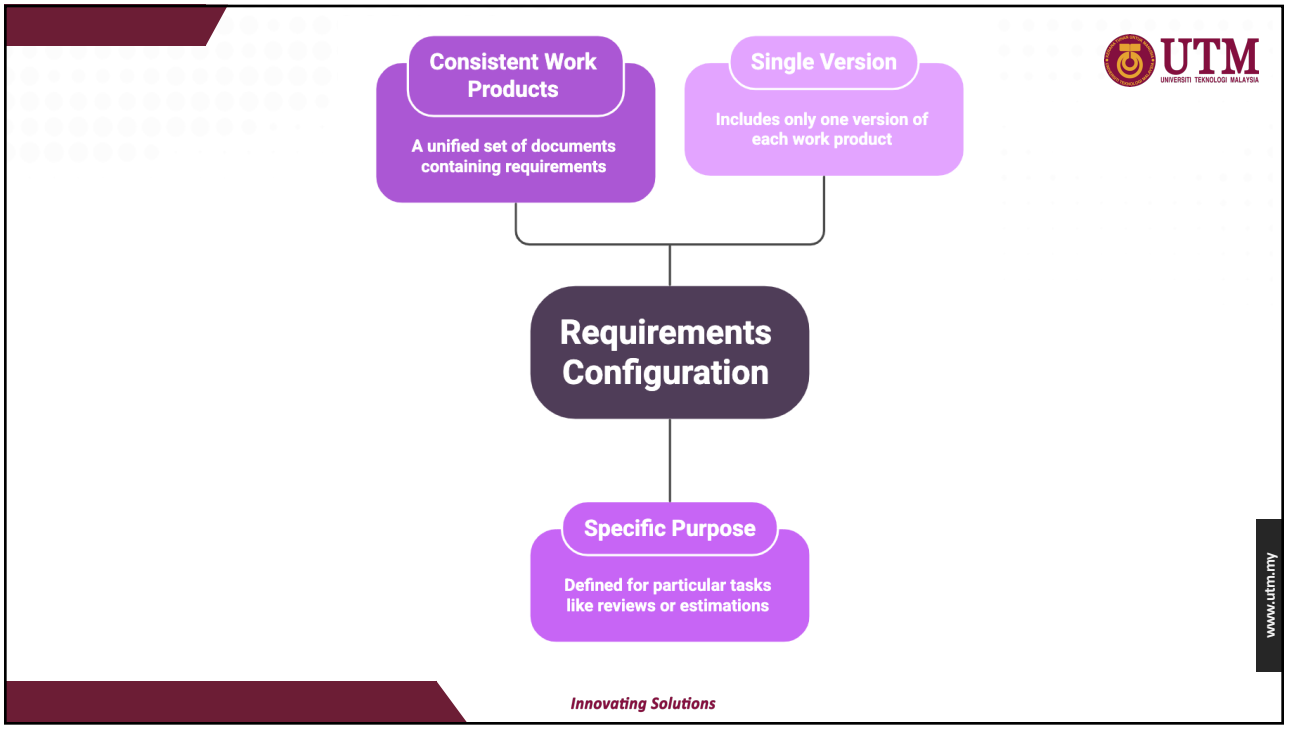
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CONFIGURATIONS & BASELINES

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Configurations



Source: Mathias Lampe (2024)

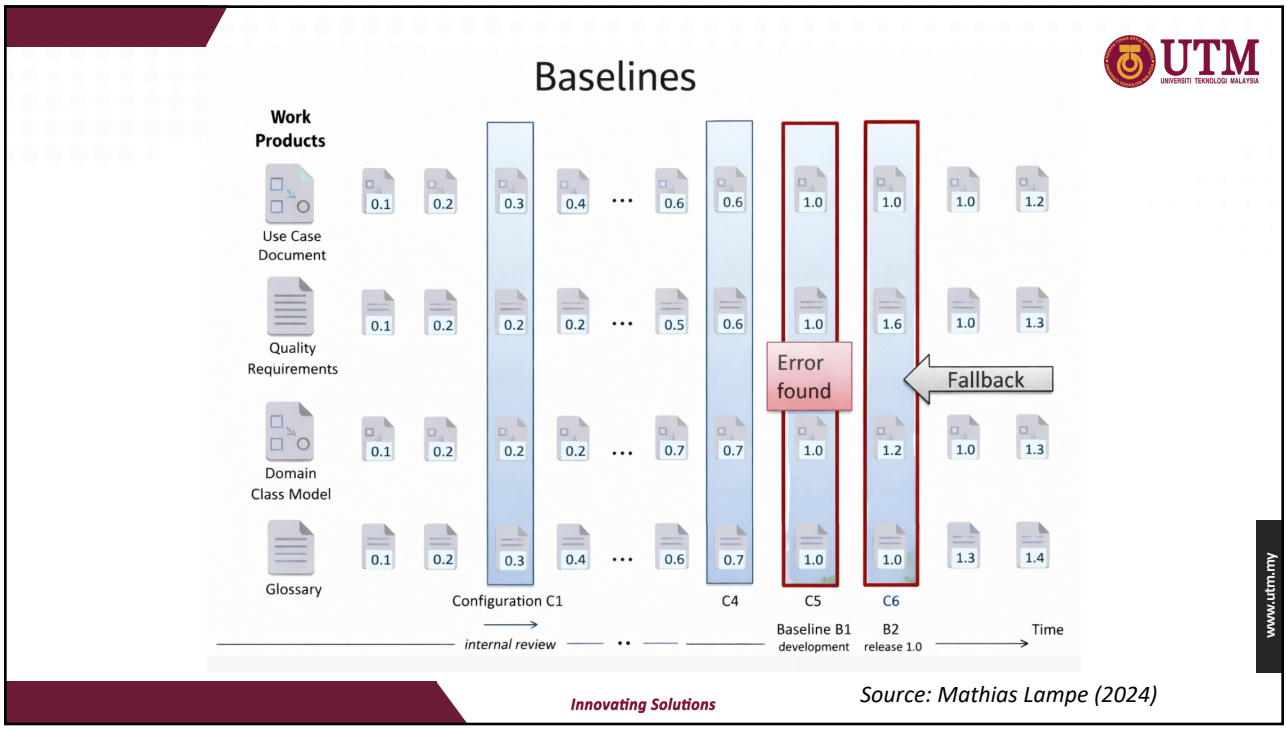
Definition: Baseline

A stable, change-controlled ↑ configuration of ↑ work products.

Note:
Baselines serve for ↑ release planning and release definition as well as for project management purposes such as effort estimation.

Definition: Requirements Baseline

A ↑ baseline for a set of ↑ requirements.



Source: Mathias Lampe (2024)

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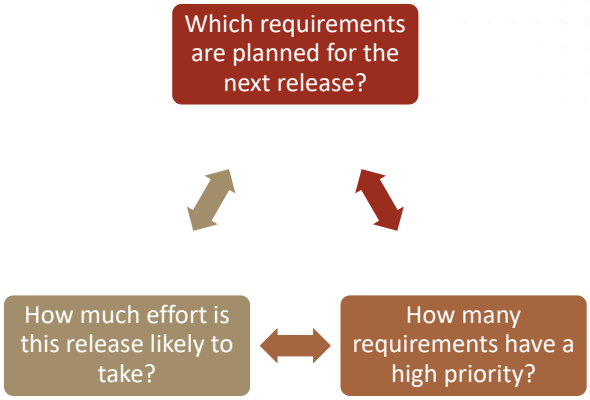


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Attributes of Requirements and Work Products



Why use Attributes?



- Enables team members and other stakeholders to get the information about the requirements they need at any point during the project
- Helps you to better manage requirements and work products

Suitable Set of Attributes

- Defining the relevant set of attributes depends on the information needs of the different stakeholders in the project
- Define set of attributes at beginning of RE in an **attribute schema**
- Existing standards, for example **ISO 29148**, provide an overview of the most relevant attributes
- Examples:
  - Identification
  - Stakeholder priority
  - Dependency
  - Risk
  - Source
  - Rationale
  - Difficulty (Effort)
  - Type

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Requirements Attributes

Attribute	Description
Identification	Unique identifier for the requirement
Stakeholder priority	Importance to relevant stakeholders
Dependency	Relationship to other requirements
Risk	Potential impact and likelihood
Source	Origin of the requirement
Rationale	Justification for the requirement
Difficulty (Effort)	Complexity and resources needed
Type	Category of the requirement

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Set of Attributes - Example

ID	Description	Rationale	Source	Type	Dependency	Priority	Risk	Difficulty
10	The system shall provide members with the ability to search items by keyword.	Library members in user group asked to do a quick search by keywords.	J. Smith	FR	11	Med.	L	M
11	The system shall provide members with the ability to browse products by categories.	Library members in user group voiced to have books and DVDs categorized to search for books.	S. Yin	FR	–	Med.	L	M
12	The system shall provide members with the ability to request borrowing all items in the item list.	Prototyping with selected library members revealed they like the concept of a "shopping cart" for borrowing multiple books.	J. Smith	FR	10, 11	High	L	H
40	The system stores all member-related data in encrypted form using the AES 4 algorithm.	One of the security best practices in the security guideline.	Security Guideline	QR	–	High	H	H
43	When the rental period expired, the system should send a reminder to the member via email.	City public library uses email reminders and has a better return rate since they introduced the feature.	S. Yin	FR	–	High	M	H
45	The system will provide library staff with the ability to view the lending statistics.	D. John voiced in interview that he and his group need statistics to determine if more books of a title need to be bought.	D. John	FR	–	Low	L	L
50	The system shall provide library staff with the ability to register a member to the system.	Library staff voiced in a workshop they prefer to register members in person rather than allow self-registration.	R. Dexter	FR	–	Med.	M	M
52	The validation transaction with the University Directory System is less than 2 seconds in 90% of the times.	One of the best practices in the usability standard.	Usability Standard	QR	–	Med.	H	M

Source: Mathias Lampe (2024)

Other useful Attributes

- Name
- Description
- Author
- Kano Model Type
- Life Cycle Status
- Planned Release

Helpful for prioritization:

- Benefit
- Loss
- Effort (Difficulty)
- Risk

Other Attributes - Example

ID	Name	Description	Rationale	Source	Author	Type	Dependency	Priority	Status	Risk	Difficulty
10	Search by keywords	The system shall provide members with the ability to search items by keywords.	Library members in user group asked to do a quick search by keywords.	J. Smith	B. Johnson	FR	11	Med.	agreed	L	M
11	Browse products	The system shall provide members with the ability to browse products by categories.	Library members in user group voiced need to browse books and DVDs.	S. Yin	A. Smith	FR	–	Med.	agreed	L	M
22	Request items	The system shall provide members with the ability to request borrowing all items in the item list.	Prototyping with selected library members revealed interest in a "shopping cart" concept.	J. Smith	A. Smith	FR	10, 11	High	suggested	L	H
40	Encrypt data	The system stores all member-related data in encrypted form.	One of the security best practices in the security guideline.	Security Guideline	B. Johnson	QR	–	High	validated	H	H
43	Check return date	When the rental period expires, the system should send a reminder to the member.	City public library uses email reminders to improve return rate.	S. Yin	A. Smith	FR	–	High	suggested	M	H
45	View lending statistics	The system will provide library staff with the ability to view lending statistics.	D. John voiced in interview the need for statistics to guide purchasing decisions.	D. John	B. Johnson	FR	–	Low	validated	L	L
50	Register member	The system shall provide library staff with the ability to register a member.	Library staff voiced in a workshop preference for in-person registration.	R. Dexter	A. Smith, B. Johnson	FR	–	Med.	agreed	M	M
52	Access time	The validation transaction with the University Directory System is less than 2 seconds in 90% of cases.	One of the best practices in the usability standard.	Usability Standard	B. Johnson	QR	–	Med.	validated	H	M

Source: Mathias Lampe (2024)

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Requirements Views

Filtering

A view filters and organizes requirements data.

Audience Specific

Shows only what a specific audience needs.

Role Support

Used to support different roles (e.g. developer, tester, project manager).

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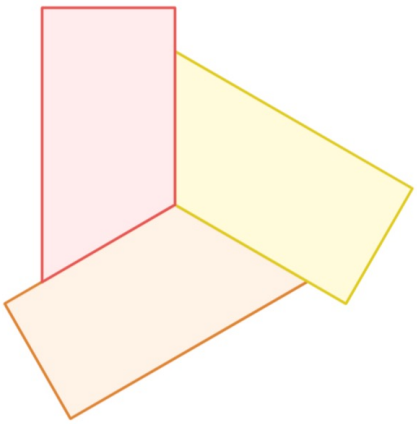
Types of Views

Aggregating View

Shows summaries such as totals, averages, and counts.

Projective View

Shows only the attributes that have been selected.



Selective View

Shows only the requirements that have been selected.

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Types of Views

Projective view (only 4 attributes)

Selective view (only sales dept)

ID	Version	Name	Type	Source	Difficulty (1...5)
1	2.0	Calculate	Functional	Sales	3
2	1.0	Response	Quality	Sales	2
3	1.2	VAT	Constraint	Sales	1
4	0.2	Foreign VAT	Constraint	Sales	4
5	2.0	Deliveryelate	Functional	Logistics	2
6	0.1	Track & trace	Functional	Logistics	5
7	1.1	Courier	Functional	Logistics	1
7	1.1	Routing	Functional	Logistics	4
9	1.2	Accessibility	Quality	Logistics	3

Aggregating view	Functional	5	Quality	2	Constraint	2

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Source: Mathias Lampe (2024)

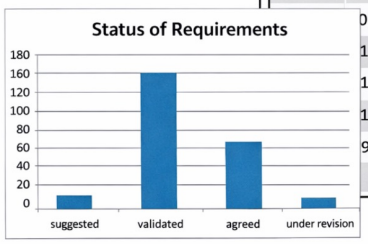
57

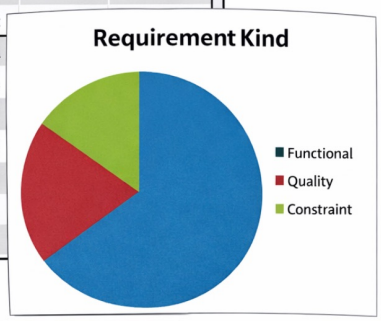
Aggregating Views

Projective view
(only 4 attributes)

Selective view
(only sales dept)

ID	Version	Name	Type	Source	Difficulty (1..5)
1	2.0	Calculate	Functional	Sales	3
2	1.0	Response	Quality	Sales	2
3	1.2	VAT	Constraint	Sales	1
4	0.2	Foreign VAT	Constraint		
5	2.0	Delivery date	Functional		
	0.1	Track & trace	Functional		
	1.1	Courier	Functional		
	1.1	Routing	Functional		
	1.2	Rouy ing	Functional		
	9	Accessibility	Quality		





Aggregating view	Functional	5	Quality	2	Constraint	2

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